# PyFluent cheat sheet

Version: 0.45.dev0

## Quick Start

```
import ansys.fluent.core as pyfluent
from ansys.units import VariableCatalog as VC
from ansys.fluent.core import (
    ScalarFieldDataRequest,
    VectorFieldDataRequest,
    SurfaceFieldDataRequest,
    SurfaceDataType)
from ansys.fluent.core.solver import *
```

## Session Management

### Launch sessions

```
solver = pyfluent.Solver.from_install()
meshing = pyfluent.Meshing.from_install()

# Connect to existing session
session = pyfluent.Solver.from_connection(ip, port,
password)
```

### File I/O

```
ReadCase(settings_source=solver)(
  file_name="<case_file>")
ReadCaseData(settings_source=solver)(
  file_name="<data_file>")

WriteCase(settings_source=solver)(
  file_name="<output_case_file>")
WriteCaseData(settings_source=solver)(
  file_name="<output_case_file>")
```

## Meshing Workflow

### Watertight geometry workflow

```
watertight = meshing.watertight()
import_geom = watertight.import_geometry
import_geom.file_name = "<geometry_file>"
import_geom.length_unit = "mm"
import_geom()

surf_mesh = watertight.create_surface_mesh
surf_mesh.cfd_surface_mesh_controls.max_size = 0.3
surf_mesh()

watertight.describe_geometry.setup_type = "fluid"
watertight.describe_geometry()

volume_mesh = watertight.create_volume_mesh_wtm
volume_mesh.volume_fill = "poly-hexcore"

volume_mesh()
```

## Physics Setup

### Boundary conditions

```
inlet = VelocityInlet(solver, name="cold-inlet")
inlet.momentum.velocity_magnitude = 0.4

turbulence = inlet.turbulence
turbulence = (turbulence.turbulence_specification.
  INTENSITY_AND_HYDRAULIC_DIAMETER)

turbulence.turbulent_intensity = 0.05
turbulence.hydraulic_diameter = "4 [in]"

inlet.thermal.temperature = 293.15
```

### Materials

```
materials = Materials(solver)
air_copy = materials.fluid.make_a_copy(from_="air",
to="air-copied")
materials.fluid["air-copied"].viscosity = 1.81e-05

# Create new material
my_solid = materials.solid.create("my-solid")
my_solid.density.value = 2650
my_solid.thermal_conductivity.value = 7.6
```

### Cell zone conditions

```
cell_zones = CellZoneConditions(solver)
cell_zones.fluid["elbow-fluid"].general.material =
"air-copied"
```

### Models

```
models = Models(solver)
models.energy.enabled = True
models.energy.viscous_dissipation = True

viscous = Viscous(solver)
viscous.model = viscous.model.K_EPSILON
k_epsilon_model = viscous.k_epsilon_model
k_epsilon_model = k_epsilon_model.REALIZABLE

rad = Radiation(solver)
rad.model = rad.model.MONTE_CARLO
rad.solve_frequency.number_of_histories = 1e7

species = Species(solver).model.option
species = species.SPECIES_TRANSPORT
```

### Solution

```
soln = Solution(solver)

methods = soln.methods
methods.p_v_coupling.flow_scheme = "Coupled"

grad_scheme = methods.spatial_discretization
  .gradient_scheme
grad_scheme = grad_scheme.GREEN_GAUSS_NODE_BASED
```

### Report definitions

```
rep_defs = ReportDefinitions(solver)
surf_rep = rep_defs.surface["outlet-temp-avg"] = {}

soln_report_type =
rep_defs.surface["outlet-temp-avg"].report_type
soln_report_type = soln_report_type.SURFACE_AREA

rep_defs.surface["outlet-temp-avg"].field = VC.TEMPERATURE
```

### Initialize and solve

```
initialization = Initialization(solver)
init_type = initialization.initialization_type
init_type = init_type.HYBRID

Initialize(solver)()

run_calc = RunCalculation(solver)
run_calc.parameters.iter_count = 100

Iterate(solver)()

Monitor(solver).residual.plot()
```

## Post-Processing

### Field data access

```
# Initialize session before requesting field data
field_data = solver.fields.field_data
abs_press_req = ScalarFieldDataRequest(
  field_name=VC.ABSOLUTE_PRESSURE,surfaces=["cold-inlet"])

abs_pressure = field_data.get_field_data(abs_press_req)

vel_req = VectorFieldDataRequest(field_name=
  VC.VELOCITY,surfaces=["cold-inlet"])
velocity = field_data.get_field_data(vel_req)

surf_req = SurfaceFieldDataRequest(data_types=[
  SurfaceDataType.Vertices, SurfaceDataType
  .FacesCentroid], surfaces=["hot-inlet", "cold-inlet"])

surfaces = field_data.get_field_data(surf_req)
vertices = surfaces["cold-inlet"].vertices
centroids = surfaces["hot-inlet"].face_centroids
```

### Field info (allowed values, active status, ranges)

```
scalar_fields = field_data.scalar_fields
vector_fields = field_data.vector_fields
surface_fields = field_data.surfaces

scalar_names = scalar_fields.allowed_values()
vector_names = vector_fields.allowed_values()
surface_names = surface_fields.allowed_values()

temp_active = scalar_fields.is_active(VC.TEMPERATURE)
vel_active = vector_fields.is_active(VC.VELOCITY)

temp_range = scalar_fields.range(VC.TEMPERATURE)
vmag_range = scalar_fields.range(VC.VELOCITY_MAGNITUDE)
```

### Reduction functions

```
reduction = solver.fields.reduction
velocity_inlet = VelocityInlet(solver, name="cold-inlet")
pressure_outlet = PressureOutlet(solver, name="outlet")

wall = WallBoundary(solver, name="wall-elbow")
area = reduction.area(locations=[velocity_inlet])
volume = reduction.volume(locations=[velocity_inlet])

avg_pressure = reduction.area_average(
  VC.ABSOLUTE_PRESSURE, locations=[pressure_outlet])
total_mass = reduction.mass_integral(
  VC.ABSOLUTE_PRESSURE, locations=[pressure_outlet])

force = reduction.force(locations=[wall])
max_vel = reduction.maximum(VC.VELOCITY_MAGNITUDE,
locations=[wall])
```

### Solution variables

```
sv_info = solver.fields.solution_variable_info
sv_data = solver.fields.solution_variable_data

zones_info = sv_info.get_zones_info()
temp_data = sv_data.get_data(variable_name=VC.TEMPERATURE,
zone_names=["elbow-fluid"], domain_name="mixture")

temp_array = sv_data.create_empty_array(VC.TEMPERATURE,
"elbow-fluid", "mixture")
temp_array[:] = 500

sv_data.set_data(variable_name=VC.TEMPERATURE,
zone_names_to_data={'elbow-fluid': temp_array},
domain_name="mixture")
```